

Juan Diego Cortés Torres | Backend Developer | .NET / C#

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🌐 <https://juandic24.github.io/portfolio-web/> • 🌐 juandic24
in [juan-diego-cortestorres](#)

Profile

Backend developer with a degree in Systems and Computer Engineering, specialized in C# (.NET) and REST API development. I have designed and deployed projects using layered and microservices architectures, JWT authentication, relational database integration, and CI/CD pipelines. I focus on writing clean, well-structured, maintainable code that is production-ready from the first commit.

Projects

Full Stack Portfolio Website (2026):

- Designed and implemented a personal website with a C#/.NET backend and a React + TypeScript frontend, integrating both layers into a single production deployment.
- Applied Tailwind CSS to build a modern, responsive, and maintainable UI without relying on external component libraries.
- Live site: <https://portfoliojdc.up.railway.app> Repository: <https://github.com/juandic24/portfolio-web>

AnimalLibrary (2026)

REST API – ASP.NET Core:

- Separated concerns across three layers (Controllers / Services / Repositories) to ensure each component has a single reason to change, making the codebase easier to maintain and test.
- Used DTOs to decouple the domain model from the presentation layer, preventing internal data from being exposed to clients.
- Implemented fully asynchronous operations with `async/await` on all PostgreSQL calls via Entity Framework Core, avoiding thread blocking under concurrent load.
- Repository: <https://github.com/juandic24/animalLibrary>

GameStore (2026)

REST Minimal API – .NET:

- Implemented JWT authentication with role-based authorization (admin/user), securing sensitive endpoints without coupling security logic to business logic.
- Applied Data Annotations validation to reject invalid input before it reaches the service layer, reducing the surface area for runtime errors.
- Covered the full resource lifecycle: create, read, update, and delete, with clearly differentiated public and protected endpoints.
- Repository: <https://github.com/juandic24/gameStore>

Microservices Application (2024)

Distributed Architecture – Backend:

- Designed each service with a single responsibility (authentication, monitoring, logging, notifications), allowing each to be deployed, scaled, and fail independently.
- Containerized all services with Docker and orchestrated them with `docker-compose`, enabling the entire distributed environment to be reproduced with a single command.
- Selected different technologies based on each service's needs (Python, JavaScript, Java), demonstrating adaptability across heterogeneous stacks.
- Repository: <https://github.com/juandic24/proyecto-final-microservicios>

PixelCNN vs. PixelCNN + VQ-VAE (2024)

Generative Model Comparison – AI / ML:

- Trained and compared two autoregressive image generation models, evaluating visual quality, training stability, and computational efficiency.
- The VQ-VAE-enhanced model showed better perceptual quality and more stable convergence, at the cost of higher implementation complexity.
- Repository: <https://github.com/juandic24/PixelCNN-PixelCNN-VQVAE-comparison>

DevOps Demo Pipeline (2026):

- Automated the full build → test → deploy cycle using a Jenkins pipeline, eliminating manual intervention on every push to the repository.
- Simulated AWS infrastructure (ECS, ECR, VPC) with LocalStack to validate the pipeline locally without incurring real cloud costs.
- Integrated Docker into the pipeline to build, tag, and deploy container images in a reproducible manner on every run.
- Repository: <https://github.com/juandic24/devops-demo>

Modern Pong (2025)

2D Game – Unity / C#:

- Built a complete Unity game with singleplayer (AI opponent) and local multiplayer modes, taking it from concept to public deployment.
- Implemented gameplay logic, UI system (menus and controls), and AI behavior for singleplayer mode, all in C#.
- Created custom pixel art visuals and audio assets.
- Play online: <https://juandic25.itch.io/modern-pong> Repository: <https://github.com/juandic24/modern-pong>

Skills

Languages: C#, Java, Python, SQL

Backend: .NET, ASP.NET Core, REST APIs, Entity Framework Core

Testing: xUnit, Moq (unit and integration testing)

Frontend: JavaScript, TypeScript, React, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, SQL Server, SQLite

DevOps: Docker, Jenkins, GitHub Actions, LocalStack

Tools: Git, GitHub, Postman

Languages: Spanish (Native), English (B2 – EF SET 62)

Education

Universidad del Quindío

B.Sc. in Systems and Computer Engineering

Colombia

2019–2025

Experience

Universidad del Quindío

Programming Teaching Assistant

Colombia

2024

- Supported students struggling with programming fundamentals, tailoring explanations to each student's individual needs and level.
- Explained data structures, algorithmic thinking, and problem-solving using Python as the reference language.
- Reinforced programming best practices and algorithmic thinking through one-on-one tutoring sessions.